

Fundamentals of Data Mining

Ensemble Learning Techniques

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Outline

- What is Ensemble Learning (EL)?
- EL Techniques and Models
- Simple
 - Majority Voting
 - Averaging
 - Weighted Averaging
- Advanced
 - Stacking and Blending
 - Bootstrap Sampling and Bagging
 - Boosting

Intuition behind Ensemble Learning

- When you want to purchase a new car, will you walk up to the nearest car shop and purchase one based on the advice of the dealer? It's highly unlikely.
- You would like to browse a few web portals where people have posted their reviews and compare different car models, checking for their features and prices. You will also probably ask your friends and colleagues for their opinion. In short, you wouldn't directly reach a conclusion, but will instead make a decision considering the opinions of other people as well.
- Ensemble models in machine learning operate on a similar idea. They combine the decisions from multiple models to improve the overall performance.

What is Ensemble Learning?

- Example - Suppose you are a movie director and you have created a short movie on a very important and interesting topic. Now, you want to take preliminary feedback (ratings) on the movie before making it public. What are the possible ways by which you can do that?
A: You may ask one of your friends to rate the movie for you.
B: Another way could be by asking 5 colleagues of yours to rate the movie.
C: How about asking 50 people to rate the movie?
- C is a better approach to get honest ratings than A and B.
- With these examples, you can infer that a diverse group of people are likely to make better decisions as compared to individuals. Similar is true for a diverse set of models in comparison to single models. This diversification in Machine Learning is achieved by a technique called **Ensemble Learning**.

EL Techniques and Models

- Simple
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EL Techniques and Models

EL Models based on Bagging and Boosting

- Bagging
 - Bagging Meta Estimator
 - Random Forest
- Boosting algorithms:
 - GBM
 - XGBoost
 - AdaBoost
 - Light GBM
 - CatBoost

Simple EL Techniques and Models

- **Majority Voting method**
 - Multiple models are used to make predictions for each data point
 - Predictions by each model are considered as a 'vote'
 - The predictions which we get from the majority of the models are used as the final prediction.
- For example, when you asked 5 of your colleagues to rate your movie (out of 5);
- The result of majority voting would be something like this:

Colleague 1	Colleague 2	Colleague 3	Colleague 4	Colleague 5	Final Rating
5	4	5	4	4	4

- Since the majority gave a rating of 4, the final rating will be taken as 4.
- This method use for **classification**
- For further reading and examples
- <https://scikit-learn.org/stable/modules/ensemble.html#voting-classifier>

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Simple EL Techniques and Models

- **Averaging**
 - Multiple predictions are made for each data point in averaging
 - Take an average of predictions from all the models and use it to make the final prediction.
- For example, the averaging method would take the average of all the values.

Colleague 1	Colleague 2	Colleague 3	Colleague 4	Colleague 5	Final Rating
5	4	5	4	4	4.4

i.e. $(5+4+5+4+4)/5 = 4.4$

- Averaging can be used for making predictions in **regression** problems

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Simple EL Techniques and Models

- **Weighted Average**
 - This is an extension of the averaging method.
 - All models are assigned different weights defining the importance of each model for prediction.
 - For instance, if two of your colleagues are critics, while others have no prior experience in this field, then the answers by these two friends are given more importance as compared to the other people.

The result is calculated as $[(5*0.23) + (4*0.23) + (5*0.18) + (4*0.18) + (4*0.18)] = 4.41$.

	Colleague 1	Colleague 2	Colleague 3	Colleague 4	Colleague 5	Final Rating
weight	0.23	0.23	0.18	0.18	0.18	
rating	5	4	5	4	4	4.41

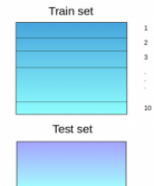
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Advanced EL Techniques and Models - Stacking

- **Stacking**
 - uses predictions from multiple models (for example decision tree, kNN, SVM) to build a new model.
 - New model is used for making predictions on the test set.

- Below is a step-wise explanation for a simple stacked ensemble:

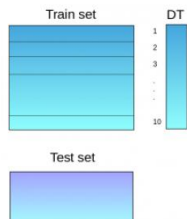
- 1. The train set is split into 10 parts.



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Advanced EL Techniques and Models - Stacking

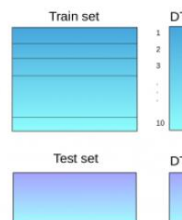
- 2. A base model (suppose a decision tree) is fitted on 9 parts and predictions are made for the 10th part. This is done for each part of the train set.



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Advanced EL Techniques and Models - Stacking

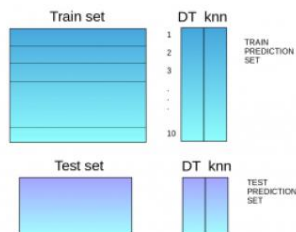
- 3. The base model (in this case, decision tree) is then fitted on the whole train dataset. Using this model, predictions are made on the test set.



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Advanced EL Techniques and Models - Stacking

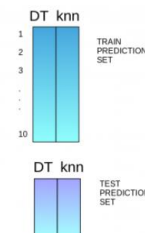
- 4. Steps 2 to 3 are repeated for another base model (say kNN) resulting in another set of predictions for the train set and test set.



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Advanced EL Techniques and Models - Stacking

- 5. The predictions from the train set are used as features to build a new model.
- 6. This model is used to make final predictions on the test prediction set.



- For further reading and examples
<https://scikit-learn.org/stable/modules/ensemble.html#stacked-generalization>

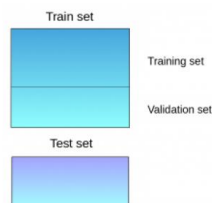
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Advanced EL Techniques and Models - Blending

- Blending
 - same approach as stacking but uses only a holdout (validation) set from the train set to make predictions.
 - In other words, unlike stacking, the predictions are made on the holdout set only.
 - The holdout set and the predictions are used to build a model which is run on the test set.

- Here is a detailed explanation of the blending process:

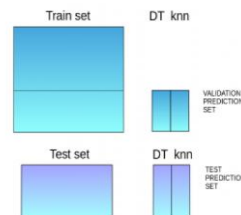
- 1. The train set is split into training and validation sets.



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Advanced EL Techniques and Models - Blending

- 2. Model(s) are fitted on the training set.
- 3. The predictions are made on the validation set and the test set.
- 4. The validation set and its predictions are used as features to build a new model.
- 5. This model is used to make final predictions on the test and meta-features.

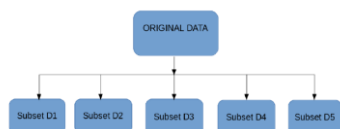


- For example and further reading-
<https://machinelearningmastery.com/blending-ensemble-machine-learning-with-python/>

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Advanced EL Techniques and Models - Bagging

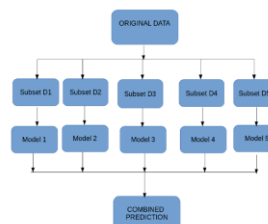
- What is Bagging**
- The idea behind bagging is combining the results of multiple models (for instance, all decision trees) to get a generalized result.
- Bootstrapping is a sampling technique in which we create subsets of observations from the original dataset, with replacement.
- Bagging (or Bootstrap Aggregating) technique uses these subsets (bags) to get a fair idea of the distribution (complete set). The size of subsets created for bagging may be less than the original set.



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Advanced EL Techniques and Models - Bagging

- 1. Multiple subsets are created from the original dataset, selecting observations with replacement.
- 2. A base model (weak model) is created on each of these subsets.
- 3. The models run in parallel and are independent of each other.
- 4. The final predictions are determined by combining the predictions from all the models.



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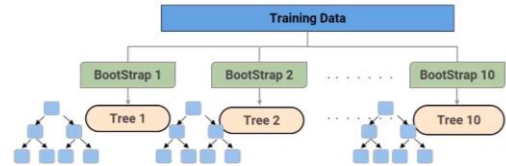
Advanced EL Techniques and Models - Bagging

- Bagging meta-estimator
- Bagging meta-estimator is an ensembling algorithm that can be used for both classification (BaggingClassifier) and regression (BaggingRegressor) problems. It follows the typical bagging technique to make predictions.
- Following are the steps for the bagging meta-estimator algorithm:
 - 1. Random subsets are created from the original dataset (Bootstrapping).
 - 2. The subset of the dataset includes all features.
 - 3. A user-specified base estimator is fitted on each of these smaller sets.
 - 4. Predictions from each model are combined to get the final result.
- For further reading and examples
<https://scikit-learn.org/stable/modules/ensemble.html#bagging-meta-estimator>

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Advanced EL Techniques and Models - Bagging

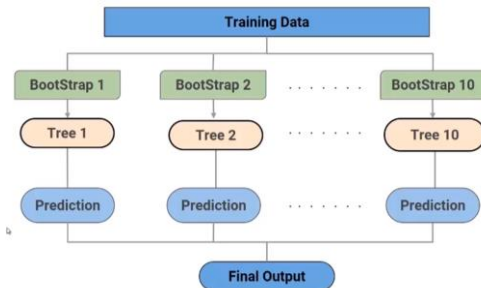
- Random Forest



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Advanced EL Techniques and Models - Bagging

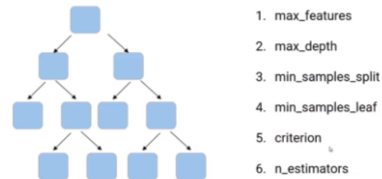
- Random Forest



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Advanced EL Techniques and Models - Bagging

- Hyperparameters of Random Forest



- For further reading and examples
<https://scikit-learn.org/stable/modules/ensemble.html#forests-of-randomized-trees>

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Advanced EL Techniques and Models - Boosting

- What is boosting
- Boosting is a sequential process, where each subsequent model attempts to correct the errors of the previous model. The succeeding models are dependent on the previous model.
- Let's understand the way boosting works in detail.
 - 1. A subset is created from the original dataset.
 - 2. Initially, all data points are given equal weights.
 - 3. A base model is created on this subset.
 - 4. This model is used to make predictions on the whole dataset.



- 5. Errors are calculated using the actual values and predicted values.

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Advanced EL Techniques and Models - Boosting

- 6. The observations which are incorrectly predicted, are given higher weights. (Here, the three misclassified blue-plus points will be given higher weights)
- 7. Another model is created, and predictions are made on the dataset. (This model tries to correct the errors from the previous model)

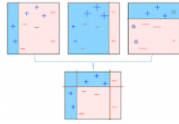


- 8. Similarly, multiple models are created, each correcting the errors of the previous model.

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Advanced EL Techniques and Models - Boosting

- 9. The final model (strong learner) is the weighted mean of all the models (weak learners).



- 10. Thus, the boosting algorithm combines a number of weak learners to form a strong learner. The individual models would not perform well on the entire dataset, but they work well for some part of the dataset. Thus, each model actually boosts the performance of the ensemble.



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Advanced EL Techniques and Models - Boosting

- For further reading and examples, <https://scikit-learn.org/stable/modules/ensemble.html#adaboost>

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Advanced EL Techniques and Models - Summary

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Advanced EL Techniques and Models

Questions



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